Lecture 13 | 20 Oct 2026

Homework #3 due Tuesday, Oct 27, 2026 (NO EXTENSIONS)

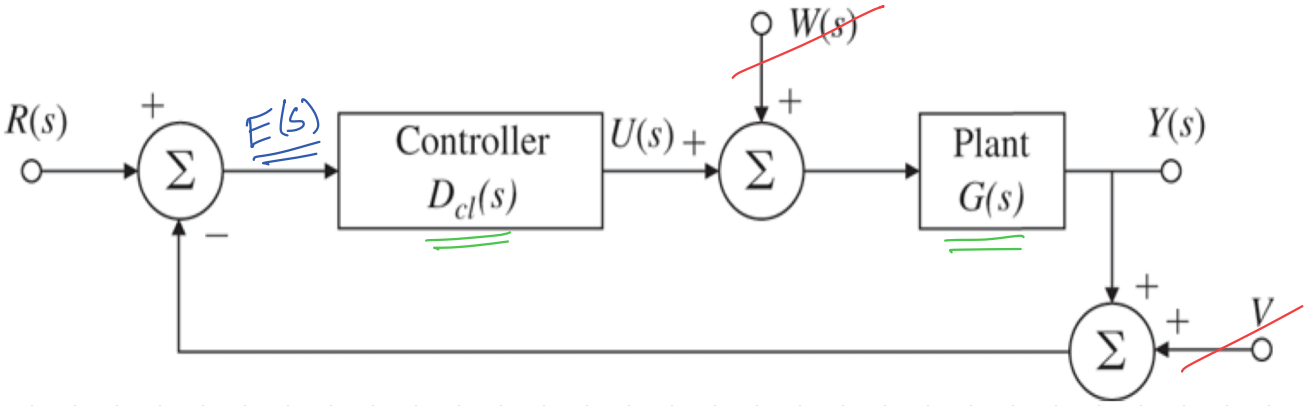
Midterm on November 5 (2.5 weeks away)

→ Covers material in FPE Ch 1-4

Homework #4 Assigned next Tuesday / due November 5

Last time: "system type" → How does a system track an input

- Consider a polynomial input
- Consider the steady state error



Combined $D_{CL}(s)$ and $G(s)$ → pulled out all of the integrators

(an integrator is $1/s^n$, where n is the order of integration

e.g. $D_{CL} = \frac{(s+1)}{s}$ $\quad G = \frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$

① Form Product

② Count the integrators

③ The Order is the system type (2 in this case)

$$e_{ss} = \lim_{s\to 0} \frac{s^n}{s^n + K_n} \cdot \frac{1}{s^k} = \begin{cases} 0 & n > k \\ \frac{1}{1+K_0} & n = k = 0 \\ \frac{1}{K_n} & n = k \geq 1 \\ \infty & n < k \end{cases}$$

$n \rightarrow$ the system type

$k \rightarrow$ the input polynomial order

$\left(r(t) = \frac{t^k}{k!} 1(t) \Longrightarrow R(s) = \frac{1}{s^{k+1}} \right)$

How do we calculate the K_n values?

$$K_p = \lim_{s\to 0} GD_{cl}(s) \quad \text{(aka } n=0\text{)} \qquad K_v = \lim_{s\to 0} s \cdot GD_{cl} \quad \text{(aka } n=1\text{)} \qquad K_a = \lim_{s\to 0} s^2 GD_{cl} \quad \text{(aka } n=2\text{)}$$

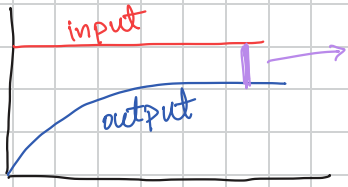
order of the input polynomial

Type	step ($k=0$)	ramp ($k=1$)	parabola ($k=2$)
Type 0	$\frac{1}{1+K_p}$	∞	∞
Type 1	0	$\frac{1}{K_v}$	∞
Type 2	0	0	$\frac{1}{K_a}$

of integrators between the controller and the plant

the steady state errors

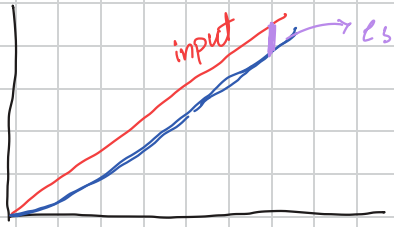
Type 0 w/ step input



$e_{ss} = \frac{1}{1+K_p}$ $\quad K_p = \lim_{s\to 0} G \cdot D_{cl}$

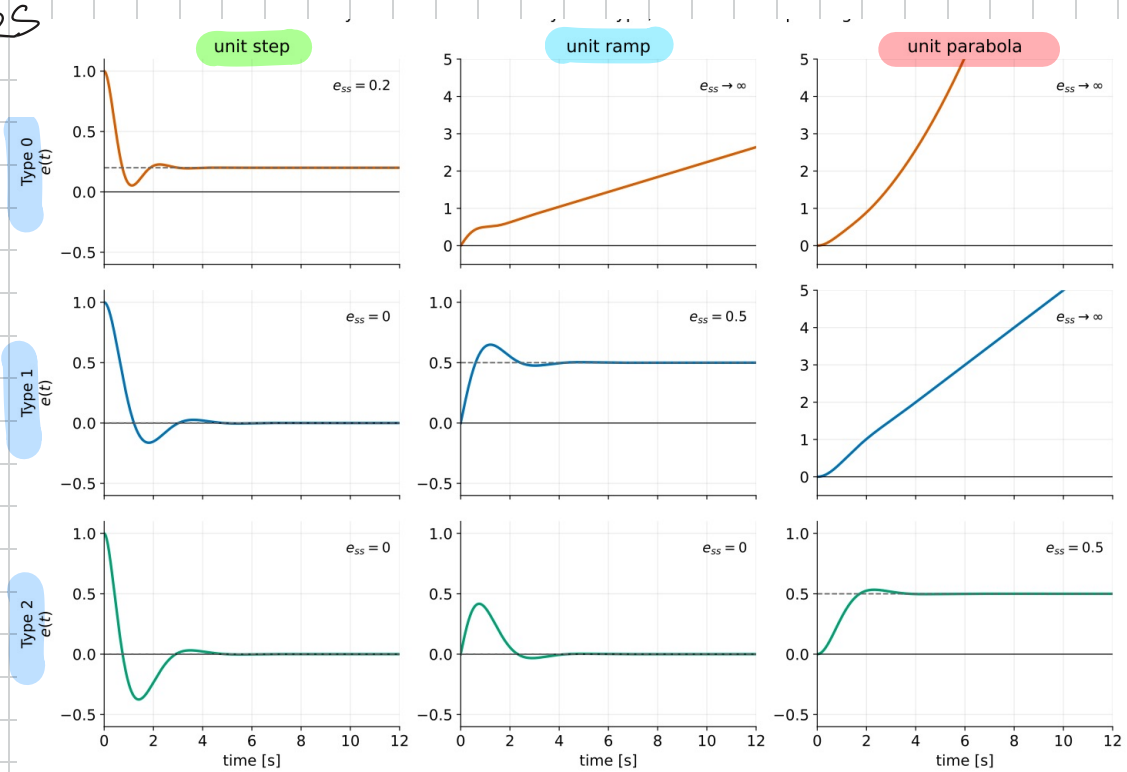
This is just a convenient trick so you don't have to manually calculate the final value theorem on the full system

Type 1 w/ Ramp Input



$e_{ss} = \frac{1}{K_v} \neq \frac{1}{1+K_1}$ $\quad K_v = \lim_{s\to 0} s \cdot GD_{cl}$

ERRORS



Type 3 → $e_{ss} \to 0$

Example: $G(s) = \frac{A}{\tau s + 1}$

P-Control: $D_{CL} = \underset{\text{lowercase}}{k_p} \Longrightarrow$ System Type is Zero

$$K_p = \lim_{s \to 0} G \cdot D_{CL} \overset{\lim s \to 0}{=} \frac{k_p \cdot A}{\tau s + 1} = k_p \cdot A$$

~~$e_{ss} \neq k_p \cdot A$~~

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + k_p \cdot A}$$ → with respect to a step input

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PI-Control: $G = \frac{A}{\tau s + 1}$ $D_{CL} = \frac{k_p s + k_I}{s}$

What is the steady state error to a step → It's zero!

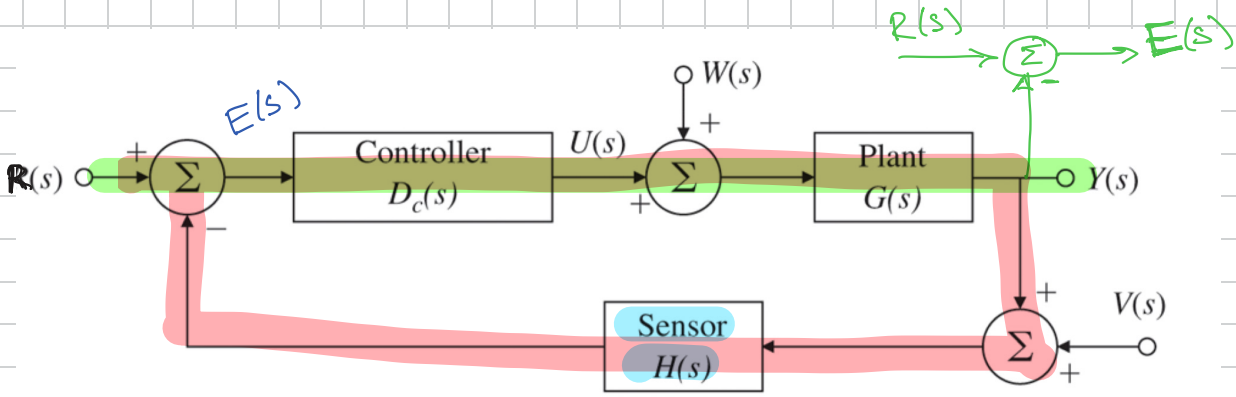
System Type = 1

$$K_v = \lim_{s \to 0} s \cdot G D_{CL} = \lim_{s \to 0} \frac{A(k_p s + k_I)}{\tau s + 1} = A k_I$$

~~$e_{ss} \neq A k_I$~~

~~$e_{ss} \neq \frac{1}{1 + A k_I}$~~

$$e_{ss} = \frac{1}{A k_I}$$ → with respect to a ramp input



Sensor Dynamics
(Low-pass to reduce noise)

system error ← E(s)
actuating error ← E(s)
} Are NOT the same errors!

System Error is what we care about b/c it's what the system is actually tracking (as opposed to the filtered output)

$$T(s) = \frac{Y}{R} = \frac{GD_c}{1 + HGD_c}$$

$$E(s) = R - Y = (1 - T(s))R \rightarrow \frac{1}{s^{k+1}} \quad \text{(polynomial of order } k\text{)}$$

How did we get this?

$Y = RT$

$$e_{ss} = \lim_{s \to 0} \frac{1 - T(s)}{s^k}$$

The filter $H(s)$ now affects the steady state error of the closed-loop

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Example $\quad G(s) = \frac{1}{s(s\tau + 1)} \quad D_c = K_p \quad H(s) = 1 + K_t s$

1: passes through
$K_t s$: scales it and takes derivative

Cruise Control w/ tachometer feedback } feedback is position + scaled velocity

$$e_{ss} = \frac{1 + K_t K_p}{K_p}$$

not a ramp input on position

$K_t \to \infty$
e_{ss} also increases

Up until this point we've considered system type with respect tracking (looking R and Y and its error)

We can also determine system type with respect to disturbances

$\Rightarrow$ Let $R = \emptyset \Rightarrow E = -Y$

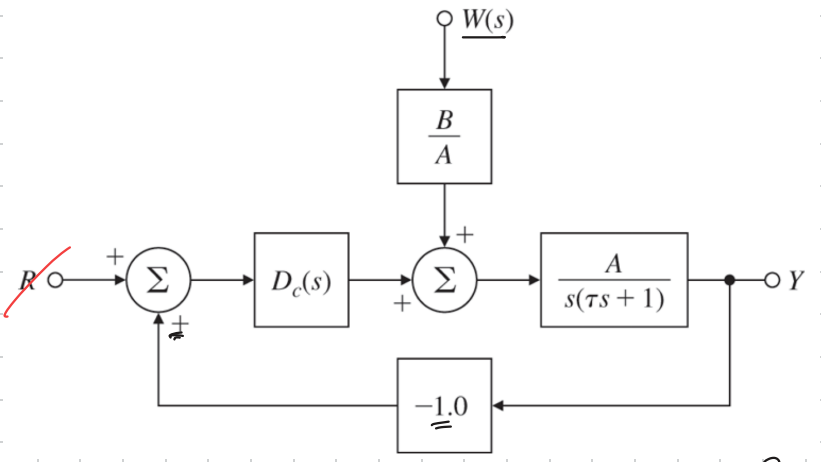
$$\frac{E}{W} = -\frac{Y}{W} = T_w(s) \quad \text{(load sensitivity)}$$

pull out all zeroes

$$e_{ss} = \lim_{s\to 0}\left(s \cdot T_w(s)\frac{1}{s^{k+1}}\right) = \lim_{s\to 0}\left(\tilde{T}_w(s)\frac{s^n}{s^k}\right)$$

input order

$$T_w = -\frac{G}{1+GD_c}$$



$$Y = \frac{A}{s(\tau s+1)}\left(-D_c Y + \frac{B}{A}W\right)$$

$$\Rightarrow \frac{Y}{W} = \frac{B}{s(s\tau+1)+AD_c}$$

$$\Rightarrow \boxed{T_w(s) = -\frac{B}{s(s\tau+1)+AD_c}}$$

Assume $D_c = K_p \Rightarrow T_w(s) = \dfrac{-B}{s(s\tau+1)+AK_p}$ (no zeroes) $\rightarrow$ Type $\emptyset$

to a step disturbance: $\boxed{e_{ss} = -\dfrac{B}{AK_p}}$

Assume $D_c = \dfrac{(K_p s + K_I)}{s} \Rightarrow T_w(s) = \dfrac{-B\cdot s}{s^2(s\tau+1)+(K_p s+K_I)A}$ (one zero) $\rightarrow$ Type 1

$\boxed{e_{ss} = \emptyset}$ to a step disturbance

$\boxed{e_{ss} = -\dfrac{B}{AK_I}}$ to a ramp disturbance

$$E = T_w(s)\cdot \underbrace{W(s)}_{=1/s} \quad \lim_{s\to 0}$$

In this case, we're not asking can the system track a reference w/ zero steady state error

Instead: we're asking can the system reject a disturbance with zero steady state error

Summary: System Type gives us a tool for determining how well a system either

① tracks a reference

② rejects a disturbance

Assumptions:

① input is a polynomial

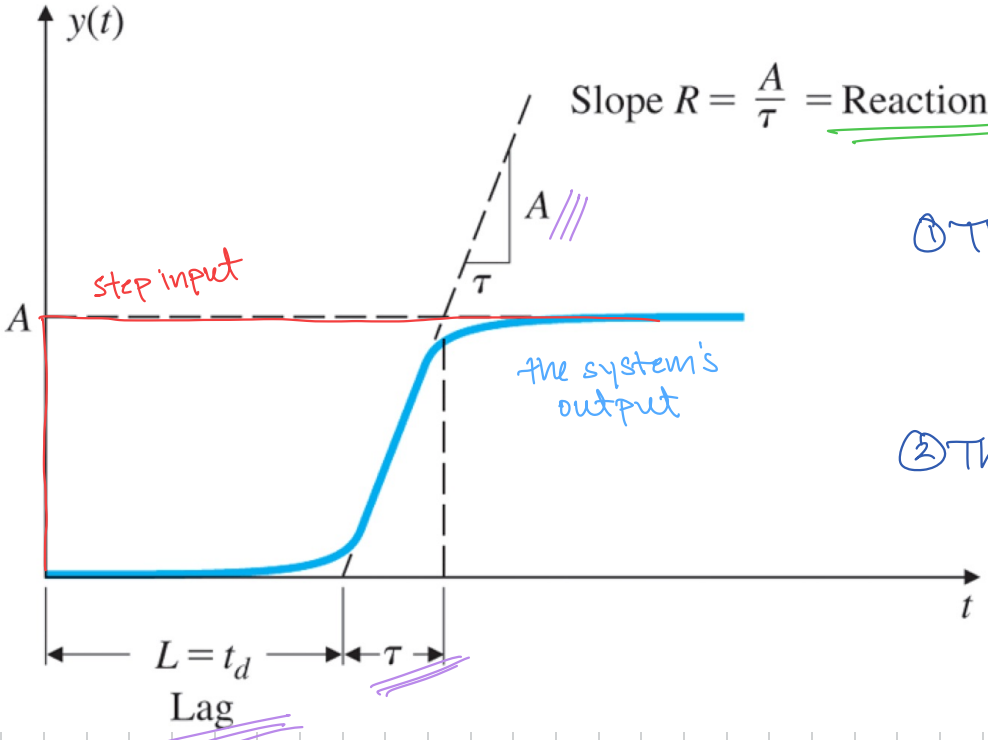
② only care about steady state

"The Error Constants": $K_P, K_V, K_a, K_j \cdots$

$$K_P = \lim_{s \to 0} G \cdot D_{cl} \qquad K_V = \lim_{s \to 0} s \cdot G \cdot D_c \cdots$$

→ Allow us to calculate steady state error given system type and input order.

PID → how to tune it (esp. from empirical data)



① This is an approximation to the real system

② The system is probably much higher order

$$\frac{Y}{U} = \frac{Ae^{-st_d}}{\tau s + 1}$$

← a delay (is an exponential in the Laplace domain)

System Identification: We command a step input and we measure the output as the above

The controller is written in the process-control (ISA "standard") form

$$D_c(s) = k_P\left(1 + \frac{1}{T_I s} + T_D s\right),$$

with $k_I = k_P/T_I$ and $k_D = k_P T_D$. Table 4.2 (tuning for a decay ratio of 0.25):

Controller	k_P	T_I	T_D
P	$\frac{1}{RL}$	–	–
PI	$\frac{0.9}{RL}$	$\frac{L}{0.3}$	–
PID	$\frac{1.2}{RL}$	$2L$	$0.5L$

PID "Tuning Rules"

Ziegler-Nichols

R and L values

Calculate K_P, T_I, T_D